

Multi-frequency synthetic CMB skies from FLAMINGO lightcones: construction, spectral validation, and tests of ILC and matched-filter cluster finding

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Abstract

We construct full-sky, multi-frequency synthetic millimetre skies from the FLAMINGO hydrodynamical lightcones of Yang et al., combining a lensed primary cosmic microwave background (CMB) with thermal and kinetic Sunyaev–Zel’dovich (tSZ, kSZ) anisotropies and the cosmic infrared background (CIB). All four components share the same lightcone rotations, so their physical cross-correlations are preserved. Maps are produced at native HEALPix resolution $N_{\text{side}} = 4096$ in thermodynamic temperature for six Planck High Frequency Instrument (HFI) channels. We specify the spectral kernels, unit conversions and frequency extensions used for each component, and the deliberate omissions (Galactic emission, radio sources, relativistic tSZ). Angular power spectra of the secondary and foreground maps recover the CIB auto- and cross-spectra and the y , kSZ and lensing auto-spectra reported by Yang et al. Applying Planck NPIPE noise and PR4 Galactic and point-source masks, we reconstruct Compton y with harmonic and needlet internal linear combination (ILC) and search for galaxy clusters with multi-frequency matched filters in iterative (iMMF) and spectrally constrained (sciMMF) modes. On a Planck-like footprint at detection signal-to-noise $q_{\text{opt}} \geq 5$, both filters achieve purity $\approx 76\text{--}77\%$ and completeness $\approx 77\%$ relative to fixed-template matched-filter truth. The mocks provide a controlled, extragalactic laboratory in which algorithm performance can be judged against known y and known clusters under realistic Planck noise and masking.

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1 Introduction

Precision tests of component separation and cluster detection at millimetre wavelengths require simulations in which the primary CMB, secondary anisotropies and extragalactic foregrounds are correlated as in the real Universe, yet the true Compton- y field and cluster population remain known. Hydrodynamical simulations that integrate tSZ, kSZ, CMB lensing and infrared emission along a common lightcone satisfy this requirement. The FLAMINGO project has released full-sky maps of these quantities from a large cosmological volume [Yang et al.(2026)], providing the physical ingredients for end-to-end Planck-like analyses.

This work builds multi-frequency temperature skies from those lightcones and uses them for two canonical applications. First, we form coadded maps with Planck beams and NPIPE instrumental noise [Planck Collaboration(2020)] and reconstruct Compton y with needlet and harmonic internal linear combination (ILC) methods following McCarthy & Hill [McCarthy & Hill(2024)]. Second, we apply multi-frequency matched filters—the iterative matched multi-filter (iMMF) and the spectrally constrained iMMF (sciMMF) of Zubeldia et al. [Zubeldia et al.(2023a), Zubeldia et al.(2023b)]—to produce cluster catalogues at detection threshold $q_{\text{opt}} \geq 5$, and measure purity and completeness against fixed-template truth signal-to-noise on the same footprint.

The sky model is intentionally minimal: lensed primary CMB, non-relativistic tSZ, kSZ and CIB, with no Galactic dust, synchrotron or free-free and no radio point sources. That choice isolates extragalactic contamination and instrumental noise as the limiting factors for SZ science, at the cost of understating foreground complexity at 545 and 857 GHz relative to flight data. Section 2 summarises the FLAMINGO inputs. Section 3 details the construction of multi-frequency temperature maps. Section 4 presents power-spectrum validation and the ILC and cluster-finding results. Section 5 states our conclusions.

2 FLAMINGO lightcone inputs

2.1 Simulation and map products

We use the flagship FLAMINGO run L2800N5040 (L2p8_m9): a comoving box of side $2.8 h^{-1} \text{Gpc}$ with 5040^3 particles per species, intermediate mass resolution m_9 , and the HYDRO_FIDUCIAL feedback calibration in a DES Year 3 $3 \times 2\text{pt} + \text{external } \Lambda\text{CDM}$ cosmology [Yang et al.(2026)]. Lightcone shells are stacked with common random rotations every box length so that cross-correlations among Compton y , kSZ, lensing convergence κ and CIB are preserved. Maps are HEALPix RING-ordered float32 fields at $N_{\text{side}} = 4096$ (pixel scale $\approx 0.86'$) in Galactic coordinates. Unless noted, we analyse a single observer (`lightcone0`).

The fields used here are:

- lensed Compton y (dimensionless);
- lensed Doppler b , related to the kSZ temperature by $\Delta T_{\text{kSZ}}/T_{\text{CMB}} = -b$;
- CMB lensing convergence κ ;
- bandpass-integrated CIB intensity at 217, 353, 545 and 857 GHz in Jy sr^{-1} , for the fiducial three-parameter dust model.

Secondary maps are already lensed shell-by-shell with the cumulative κ field [Yang et al.(2026)]. The cosmology adopted for the primary CMB matches Table 1 of Yang et al.:

$$h = 0.681, \Omega_{\text{m}} = 0.306, \Omega_{\text{b}} = 0.0486, m_{\nu} = 0.06 \text{ eV}, A_{\text{s}} = 2.099 \times 10^{-9}, n_{\text{s}} = 0.967. \quad (1)$$

We use $T_{\text{CMB}} = 2.7255 \text{ K}$ consistently for all spectral conversions.

2.2 Scope of the mock sky

The additive sky model has exactly four components,

$$\Delta T(\hat{n}, \nu) = \Delta T_{\text{CMB}}(\hat{n}) + \Delta T_{\text{tSZ}}(\hat{n}, \nu) + \Delta T_{\text{kSZ}}(\hat{n}) + \Delta T_{\text{CIB}}(\hat{n}, \nu), \quad (2)$$

all expressed in thermodynamic μK_{CMB} after spectral conversion. We deliberately exclude:

- Galactic emission (thermal dust, synchrotron, free-free);
- radio point sources present in the FLAMINGO release;
- relativistic tSZ corrections;
- star-formation-rate maps as an additive temperature term (they are precursors of the CIB, not a separate sky component);
- κ as a temperature term (κ enters only through lensing of the primary CMB; secondary maps are already lensed upstream).

Instrumental noise is not baked into the component maps; it is added at analysis time from official Planck NPIPE residual realisations [Planck Collaboration(2020)]. Halo lightcone catalogues provide truth positions, redshifts and masses for cluster validation.

3 Construction of multi-frequency temperature maps

3.1 Primary CMB and lensing

An unlensed Gaussian temperature realisation is drawn from the unlensed scalar C_ℓ^{TT} spectrum computed with CAMB [Lewis & Challinor(2006)] at the FLAMINGO cosmology (units μK^2). The FLAMINGO κ map is mean-subtracted and converted to lensing-potential harmonics,

$$\phi_{\ell m} = \frac{2}{\ell(\ell+1)} \kappa_{\ell m} \quad (\ell \geq 2), \quad (3)$$

with monopole and dipole of ϕ set to zero. Deflection is performed on a full-sky CAR grid at resolution $\pi/(2N_{\text{side}})$ and reprojected to HEALPix. The result is a lensed primary CMB map at $N_{\text{side}} = 4096$ in μK_{CMB} .

3.2 Thermal Sunyaev–Zel’dovich effect

Given the lensed Compton- y map, the non-relativistic thermodynamic temperature at frequency ν is

$$\Delta T_{\text{tSZ}}(\nu) = T_{\text{CMB}} y f(x), \quad x = \frac{h\nu}{k_{\text{B}} T_{\text{CMB}}}, \quad f(x) = x \coth\left(\frac{x}{2}\right) - 4. \quad (4)$$

We evaluate Equation (4) at the six Planck HFI frequencies $\{100, 143, 217, 353, 545, 857\}$ GHz. The dimensionless y map is retained as ground truth for ILC and cluster studies.

3.3 Kinetic Sunyaev–Zel’dovich effect

In thermodynamic units the kSZ is frequency-independent:

$$\Delta T_{\text{kSZ}} = -T_{\text{CMB}} b. \quad (5)$$

A single ΔT map therefore serves all channels.

3.4 Cosmic infrared background

Released CIB maps at 217, 353, 545 and 857 GHz are bandpass-convolved specific intensities I_ν in Jy sr^{-1} . Conversion to thermodynamic temperature uses

$$\Delta T_{\text{CIB}}(\nu) = \frac{I_\nu}{dB_\nu/dT|_{T_{\text{CMB}}}} \times 10^6 \mu\text{K}. \quad (6)$$

Frequencies outside the released set are approximated as follows [Yang et al.(2026)]: exact released bands are used unchanged; intermediate frequencies use logarithmic interpolation in I_ν ; frequencies below 217 GHz (including 100 and 143 GHz) are scaled from the nearest released band with the three-parameter greybody SED at effective redshift $z_{\text{eff}} = 1.5$,

$$\Theta(\nu, T_d) \propto \frac{\nu^{\beta_d+3}}{e^{h\nu/k_B T_d} - 1}, \quad T_d(z) = T_0(1+z)^\alpha, \quad (7)$$

with $\beta_d = 1.65$, $T_0 = 35.14 \text{ K}$ and $\alpha = 0$. This approximation neglects frequency decorrelation of small-scale structure, which is acceptable where the CIB is subdominant to the CMB and instrumental noise.

3.5 Beams, noise and masks

Beams. Planck HFI Gaussian full-width half-maxima (instrument paper Table I) are $\{9.66', 7.22', 4.90', 4.92', 4.92', 4.92'\}$ at $\{100, 143, 217, 353, 545, 857\}$ GHz. Component products are stored beam-unconvolved; beams are applied when forming total skies for ILC and matched filters.

Noise. Planck NPIPE noise Monte Carlos are end-to-end residual maps (pipeline output minus input), HEALPix $N_{\text{side}} = 2048$, full sky, anisotropic and position-tied [Planck Collaboration(2020)]. Channels at 100–353 GHz are in K_{CMB} ; 545 and 857 GHz are in MJy sr^{-1} and require conversion through dB_ν/dT . Independently destriped detector-set splits A and B provide noise-decoupled cross-spectra. Noise must not be rotated relative to the signal.

Masks. PR4 NILC multi-field masks supply a soft Galactic plane cut and binary point-source holes. Analyses use a binary $\text{GAL} \times \text{PS}$ product at $N_{\text{side}} = 2048$. Matched-filter tiling further apodises flat-sky galaxy, point-source and tile-boundary masks.

3.6 Analysis applications

ILC Compton- y reconstruction. Multi-frequency coadds (Equation 2) are beam-smoothed, downgraded to $N_{\text{side}} = 2048$ and combined with NPIPE splits. Harmonic ILC (HILC) and needlet ILC (NILC) preserve the tSZ spectral response, optionally deprojecting CMB and/or CIB [McCarthy & Hill(2024)]. Spectra are reported after deconvolution of the common $5'$ working beam for comparison with beam-free truth y .

Matched-filter cluster finding. SZiFi implements iterative multi-frequency matched filters (iMMF) and spectrally constrained filters that null a CIB spectral template (sciMMF) [Zubeldia et al.(2023a), Zubeldia et al.(2023b)]. The sky is analysed in 14.8° tiles with NPIPE noise and PR4 masks; catalogues are thresholded at $q_{\text{opt}} > 5$. Truth signal-to-noise for completeness uses a fixed-template matched filter on the known y field within the same footprint and redshift cut.

4 Results

4.1 Component maps

Figure 1 shows the four components at 353 GHz. Figure 2 highlights the lensed primary CMB and the full-sky Compton- y field. Frequency dependence of tSZ and CIB is shown in Figures 3 and 4. The maps exhibit the expected morphology: acoustic structure in the CMB, cluster-scale peaks in y , a diffuse kSZ pattern, and dusty large-scale structure in the CIB that brightens toward higher frequencies.

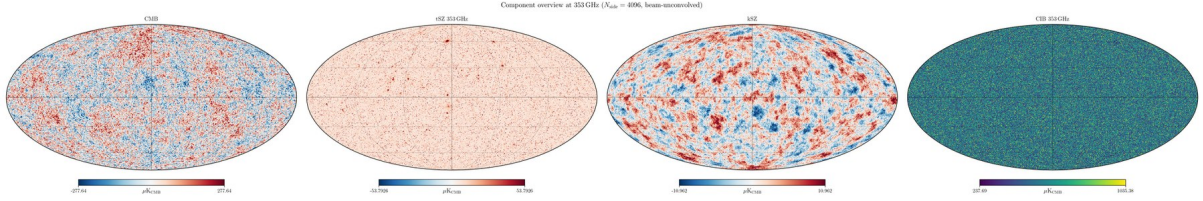


Figure 1: Four-component sky model at 353 GHz (Mollweide, Galactic coordinates, $N_{\text{side}} = 4096$). Clockwise from top-left: lensed primary CMB, tSZ ΔT , kSZ ΔT , and CIB ΔT (all temperatures in μK_{CMB}).

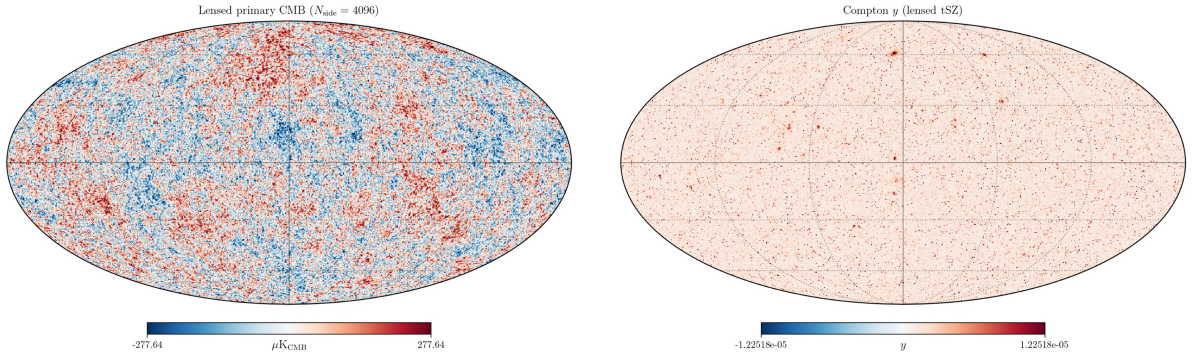


Figure 2: Lensed primary CMB (left; μK_{CMB}) and Compton- y ground truth (right).

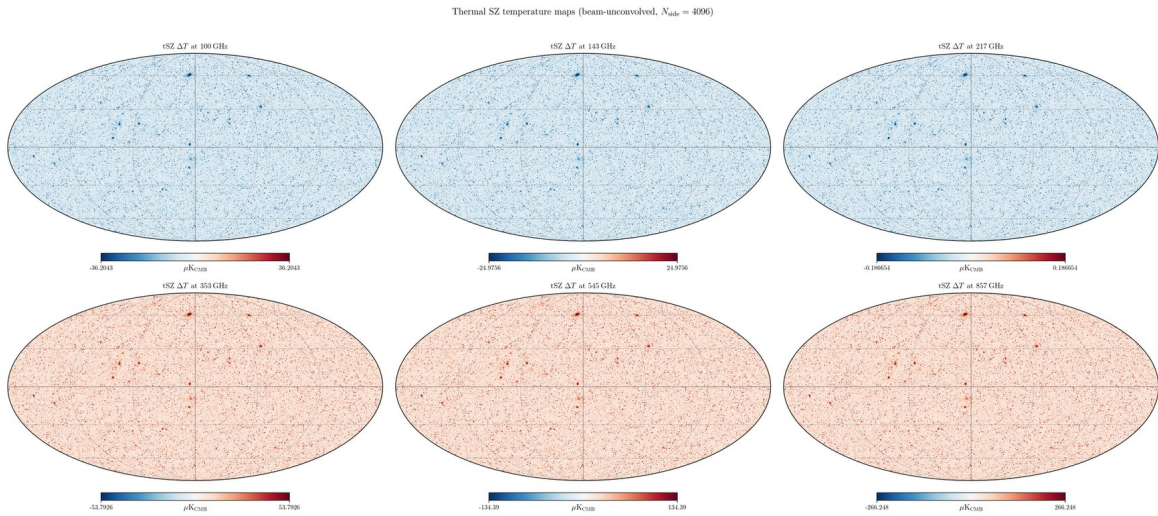


Figure 3: tSZ thermodynamic temperature at the six Planck HFI frequencies (Equation 4). The spectral null near 217 GHz and the sign change across it are visible.

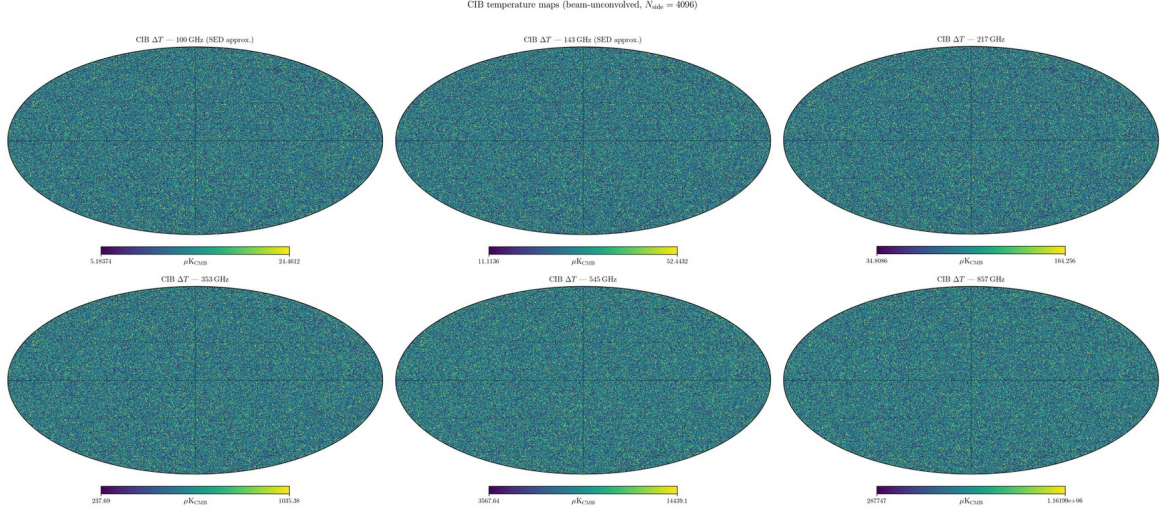


Figure 4: CIB in μK_{CMB} after conversion from released intensity maps, with greybody SED scaling at 100 and 143 GHz.

4.2 Power-spectrum validation

We recompute angular power spectra of the native-resolution integrated maps with monopole subtraction and HEALPix pixel-window deconvolution. Multipoles are binned with $\Delta\ell = 7$ following Yang et al.

Figure 5 shows CIB auto-spectra at 217–857 GHz. Figure 6 reports the mean frequency decorrelation

$$r_{\nu\nu'} = \frac{C_{\ell}^{\nu \times \nu'}}{\sqrt{C_{\ell}^{\nu\nu} C_{\ell}^{\nu'\nu'}}} \quad (8)$$

averaged over $150 < \ell < 1000$, reproducing the structure of Yang et al. Table 3: strong correlation among neighbouring bands and progressive loss of coherence between 217 and 857 GHz. Figure 7 shows CIB $\times y$ and CIB $\times \kappa$ cross-spectra. Figure 8 shows the Compton- y , kSZ (thermodynamic) and κ auto-spectra. These measurements confirm that the lightcone products and unit conversions recover the published secondary and foreground statistics at $N_{\text{side}} = 4096$.

4.3 Instrumental noise

Figure 9 shows NPIPE detector-set A residual maps at 100–353 GHz for one Monte Carlo realisation. Depth varies strongly across the sky (ecliptic-plane striping and polar deep fields), which motivates split-based cross-spectra and footprint cuts in both ILC and cluster finding.

4.4 ILC reconstruction of Compton- y

4.4.1 Baseline HILC and NILC

Figure 10 compares the truth y auto-spectrum with beam-deconvolved HILC ($A \times B$) and NILC reconstructions on coadds that include NPIPE noise. Figure 11 shows the transfer $T_{\ell} = C_{\ell}^{y_{\text{ILC}} \times y_{\text{true}}} / C_{\ell}^{yy}$. Both methods recover the large-scale y power; residuals at high multipoles reflect noise and residual foregrounds after beam deconvolution.

4.4.2 Constrained ILC deprojection suite

Following McCarthy & Hill [McCarthy & Hill(2024)], we run a suite of *constrained* (deprojected) HILC reconstructions that preserve unit response to tSZ while nulling one or more foreground

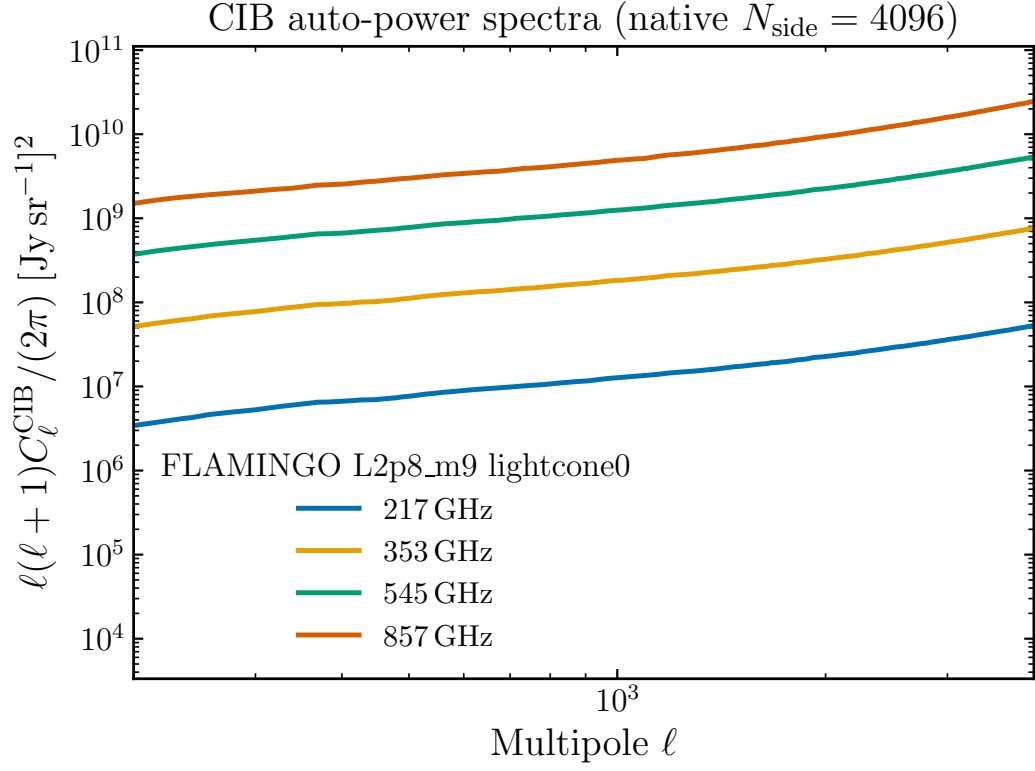


Figure 5: CIB auto-power spectra at $N_{\text{side}} = 4096$ (cf. Yang et al. Fig. 8, left).

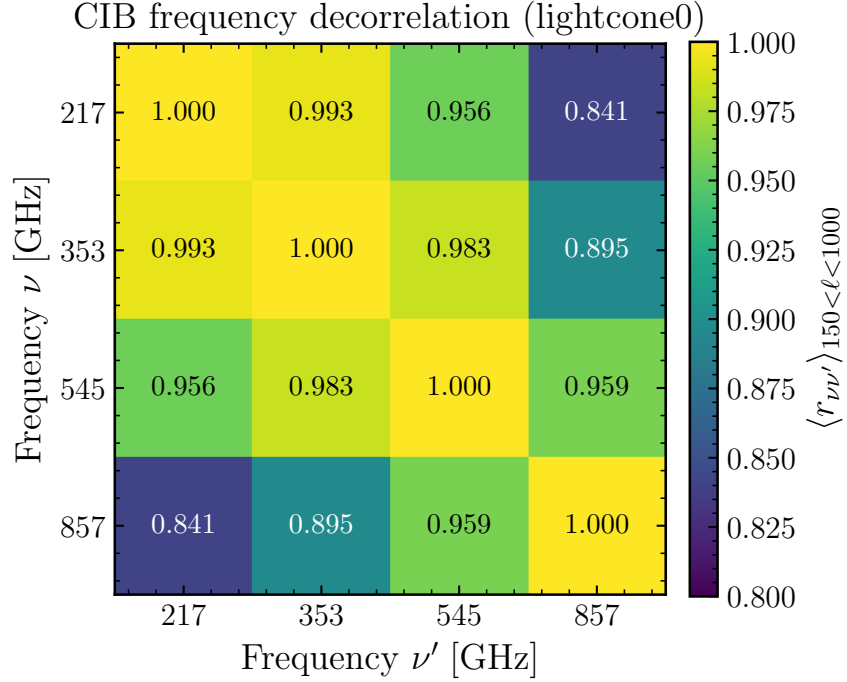


Figure 6: Mean CIB frequency decorrelation $\langle r_{\nu\nu'} \rangle$ for $150 < \ell < 1000$ (cf. Yang et al. Table 3).

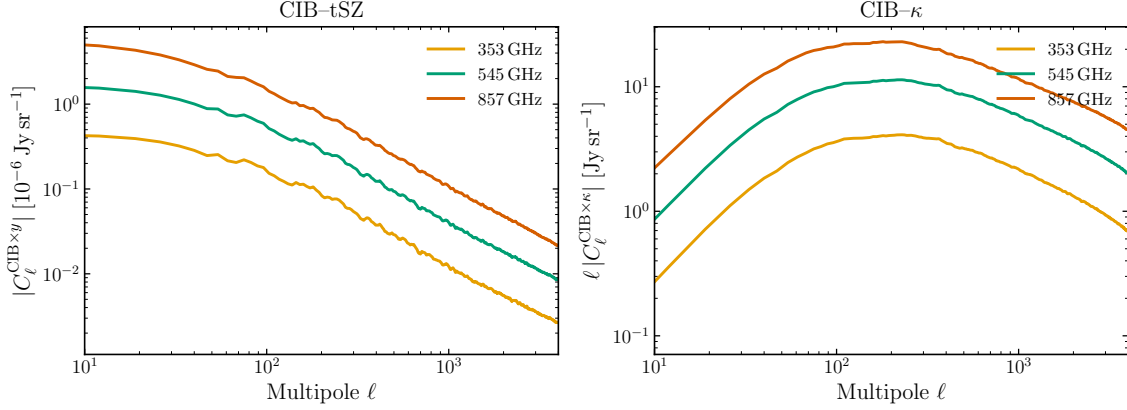


Figure 7: Absolute $\text{CIB} \times y$ (left) and $\ell |C_\ell^{\text{CIB} \times \kappa}|$ (right) at 353, 545 and 857 GHz (cf. Yang et al. Fig. 10).

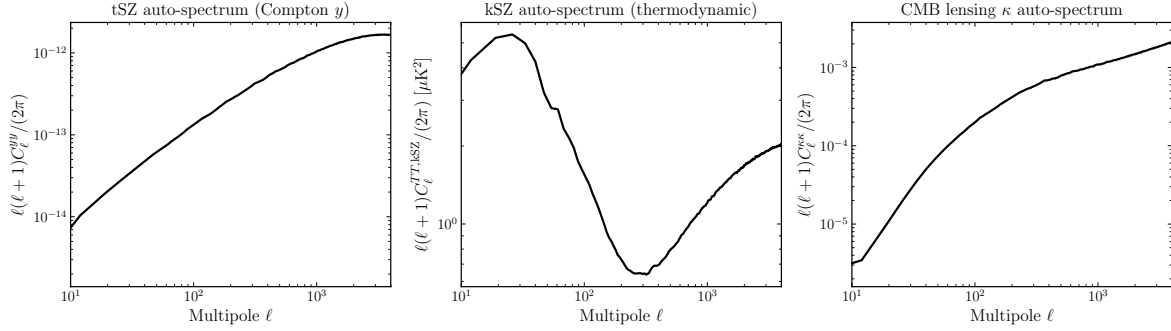


Figure 8: Auto-spectra of Compton y (left), kSZ thermodynamic temperature (centre) and CMB lensing κ (right) at native resolution.

spectral energy distributions (SEDs). With six HFI channels the available options include nulling the CMB blackbody, a modified-blackbody CIB template, and first-order CIB SED derivatives with respect to spectral index ($\delta\beta$) and dust temperature (δT). The mock sky contains CMB, tSZ, kSZ and CIB only, so CIB deprojection is the primary science-motivated constraint; CMB deprojection also removes the thermodynamic kSZ because the two share a blackbody SED in μK_{CMB} .

Table 1 summarises validation metrics against the beam-convolved truth y map: map standard deviation σ_y , full-sky correlation after common-beam matching, mid- ℓ correlation ρ_{50-500} , and a mid- ℓ transfer estimate T_{mid} . Figure 12 visualises the same suite; Figure 13 shows auto-power spectra for selected deprojections.

The results encode a clear bias–noise trade-off. **CIB-only** deprojection raises ρ_{50-500} from 0.654 to 0.671 and T_{mid} from 0.909 to 0.960 with only a modest increase in σ_y . Adding CMB to the constraint set leaves metrics essentially unchanged, as expected on a mock without Galactic dust. First-order $\delta\beta$ deprojection already reduces mid- ℓ correlation, and the full moment set CIB + $\delta\beta$ + δT (with or without CMB) more than doubles σ_y and collapses ρ_{50-500} to ~ 0.27 . On this Galactic-free mock, therefore, mild CIB deprojection is beneficial for correlation with truth y , whereas aggressive multi-moment constraints are noise-dominated—consistent with the behaviour reported for Planck PR4 in [McCarthy & Hill(2024)].

Planck NPIPE noise realisations (detector-set A)

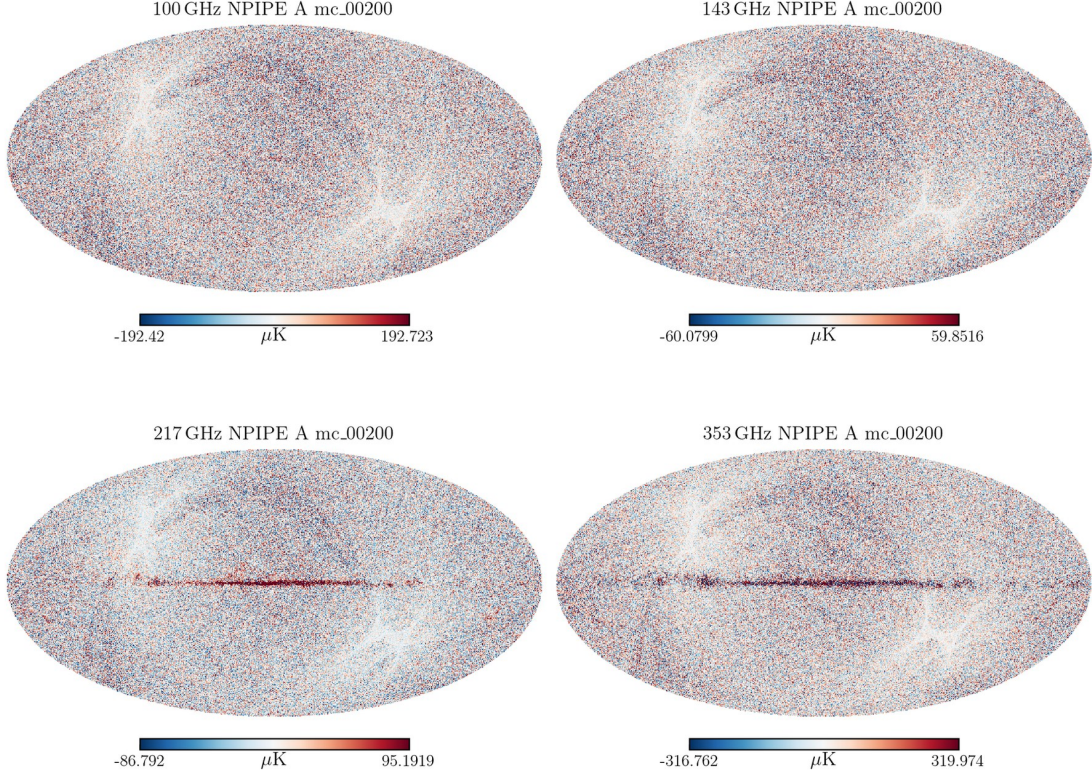


Figure 9: Planck NPIPE noise realisation mc_00200, detector-set A, at 100, 143, 217 and 353 GHz (μK_{CMB}).

Table 1: HILC constrained-ILC deprojection suite (NPIPE split A) versus truth Compton- y . Mild CIB deprojection improves mid- ℓ correlation and transfer; multi-moment CIB constraints ($\delta\beta + \delta T$) inflate the residual variance by a factor of ~ 2 –3 and destroy correlation with truth.

Deprojection	σ_y	corr (beamed)	ρ_{50-500}	T_{mid}
None	1.81×10^{-6}	0.473	0.654	0.909
CMB	1.82×10^{-6}	0.470	0.655	0.910
CIB	1.84×10^{-6}	0.496	0.671	0.960
CIB+CMB	1.85×10^{-6}	0.493	0.672	0.959
CIB+ $\delta\beta$	1.95×10^{-6}	0.463	0.614	0.949
CIB+ $\delta\beta$ +CMB	1.97×10^{-6}	0.458	0.615	0.948
CIB+ $\delta\beta$ + δT	4.19×10^{-6}	0.226	0.279	1.000
CIB+ $\delta\beta$ + δT +CMB	6.00×10^{-6}	0.159	0.271	0.995

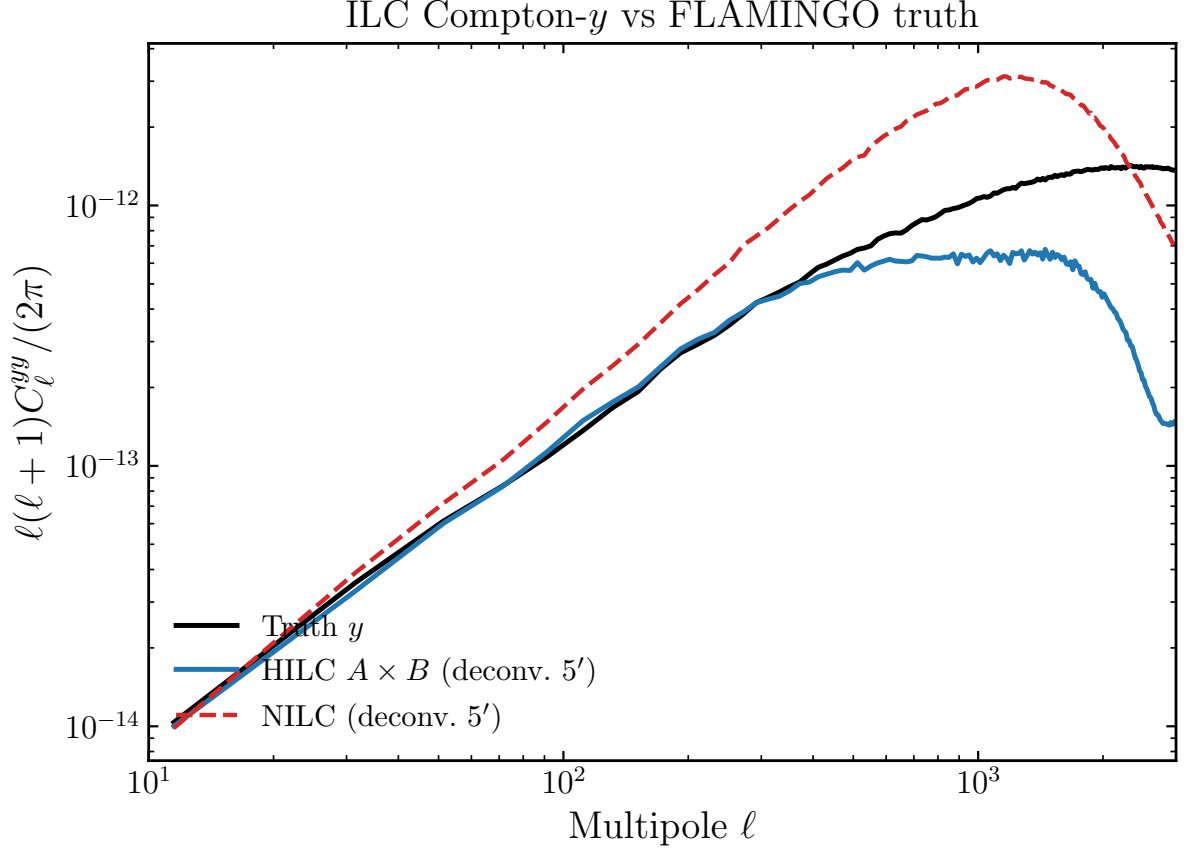


Figure 10: Beam-deconvolved Compton- y power spectra: FLAMINGO truth versus HILC (split $A \times B$) and NILC on mock skies with NPIPE noise.

4.5 Matched-filter cluster catalogues

We run iMMF and sciMMF (CIB spectral constraint) on multi-frequency tiles with NPIPE split A, PR4 GAL $\times$ PS masking, and catalogue threshold $q_{\text{opt}} \geq 5$. Detections are matched to truth systems with $q_{\text{true}} \geq 5$ (fixed-template matched filter on truth y) within $10'$, restricted to $z \leq 1$ and the unmasked footprint.

4.5.1 Sky maps of found clusters

Figure 14 shows all iMMF detections with $q_{\text{opt}} \geq 5$ overlaid on the truth Compton- y map inside the PR4 footprint; circles scale with the fitted θ_{500} . Figure 15 is the corresponding sciMMF catalogue. Figures 16 and 17 colour the same detections by match outcome: lime true positives and red false positives under the $10'$ Zubeldia-style match to fixed-MMF truth. High- q systems concentrate on bright y peaks; residual false positives include both threshold noise fluctuations and sources near the matching boundary.

4.5.2 Purity and completeness

Table 2 summarises the catalogues. Both methods achieve purity (≈ 76 – 77%). Completeness relative to the fixed-MMF-detectable truth sample is $\approx 77\%$ for both; sciMMF returns more detections (832 versus 731 after footprint cuts) at essentially the same purity. Figure 18 shows purity and completeness summary bars (completeness numerators count detectable truths that were found: 333/430 for iMMF and 334/430 for sciMMF); Figure 19 shows completeness versus

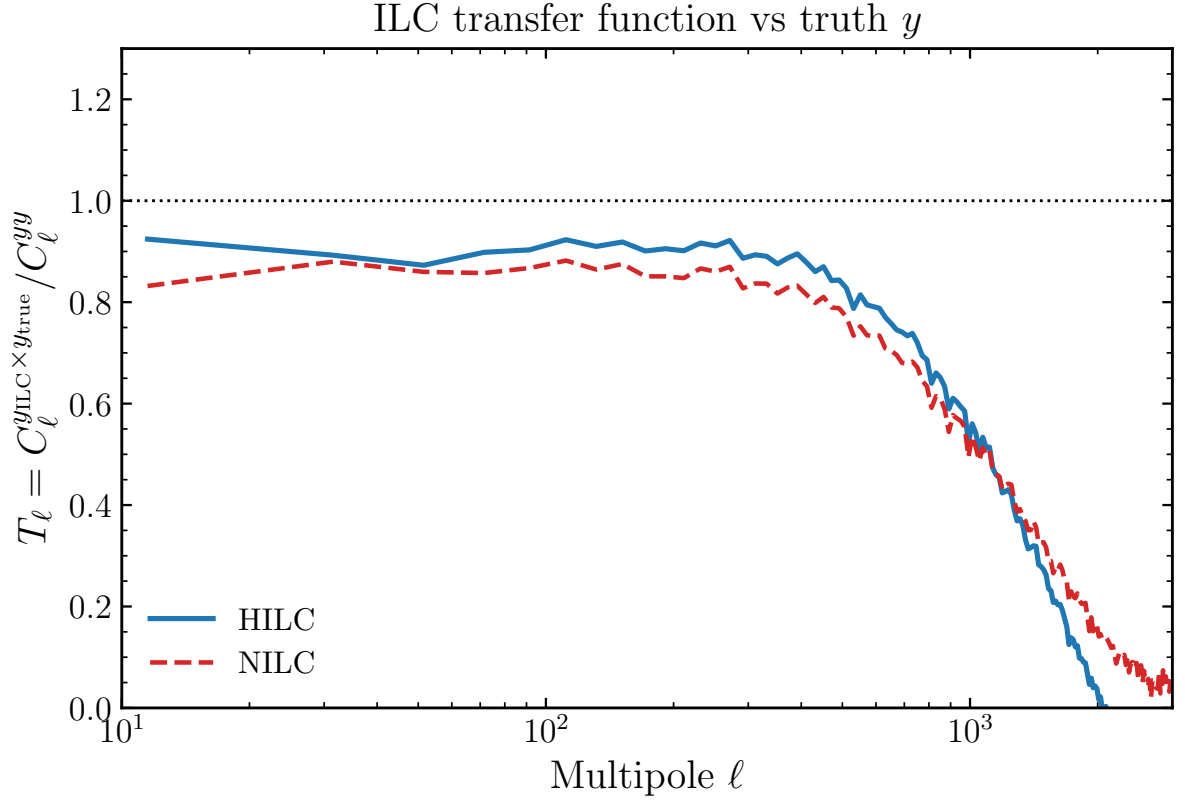


Figure 11: ILC transfer function relative to truth y for HILC and NILC.

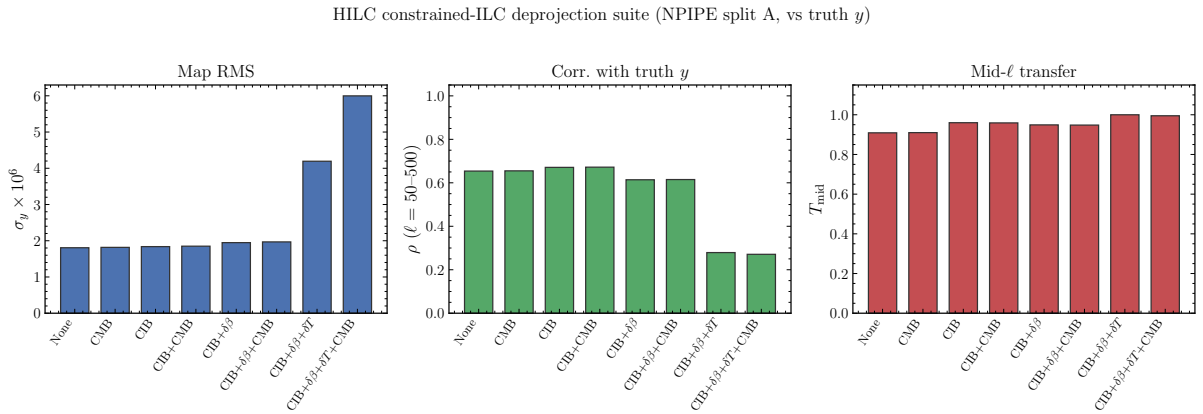


Figure 12: HILC deprojection suite metrics versus truth y : map RMS (left), multipole-windowed correlation ρ_{50-500} (centre), and mid- ℓ transfer (right).

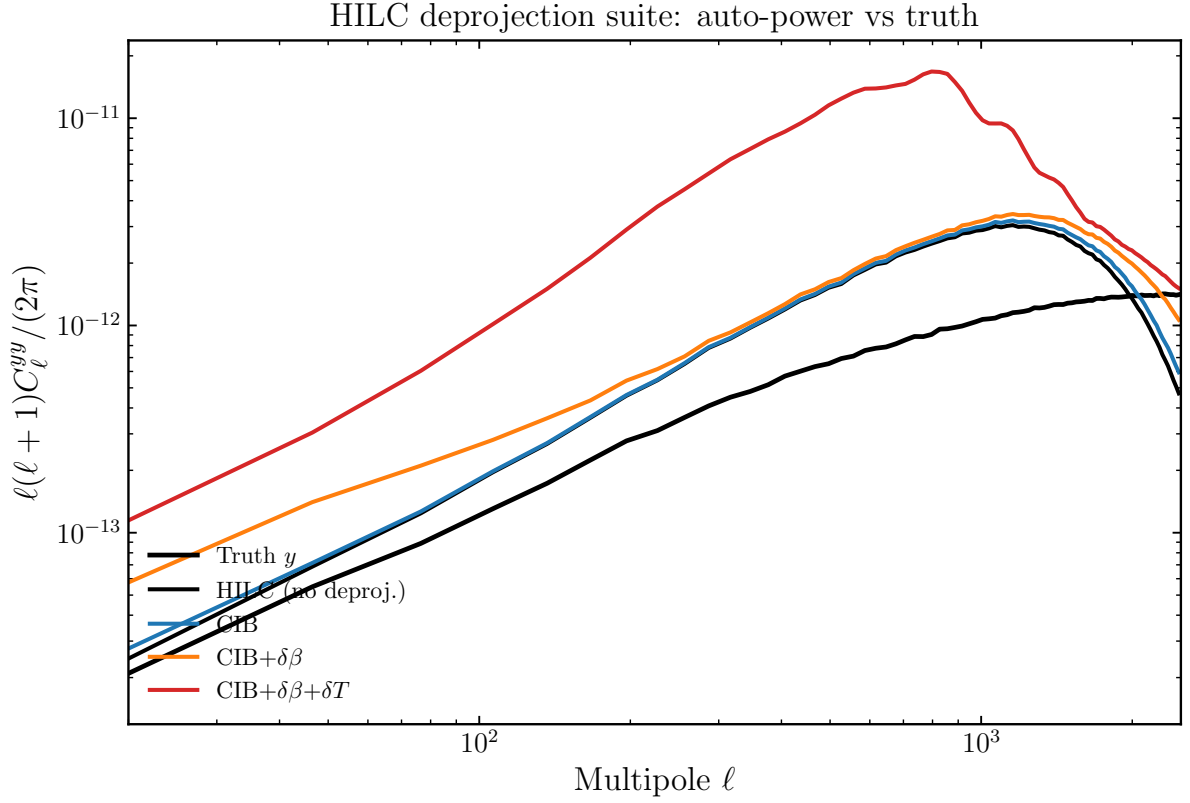


Figure 13: Beam-deconvolved auto-power for selected HILC deprojections compared with truth y . CIB-only deprojection stays close to the unconstrained solution; adding $\delta\beta + \delta T$ moments raises small-scale power (noise penalty) by a large factor.

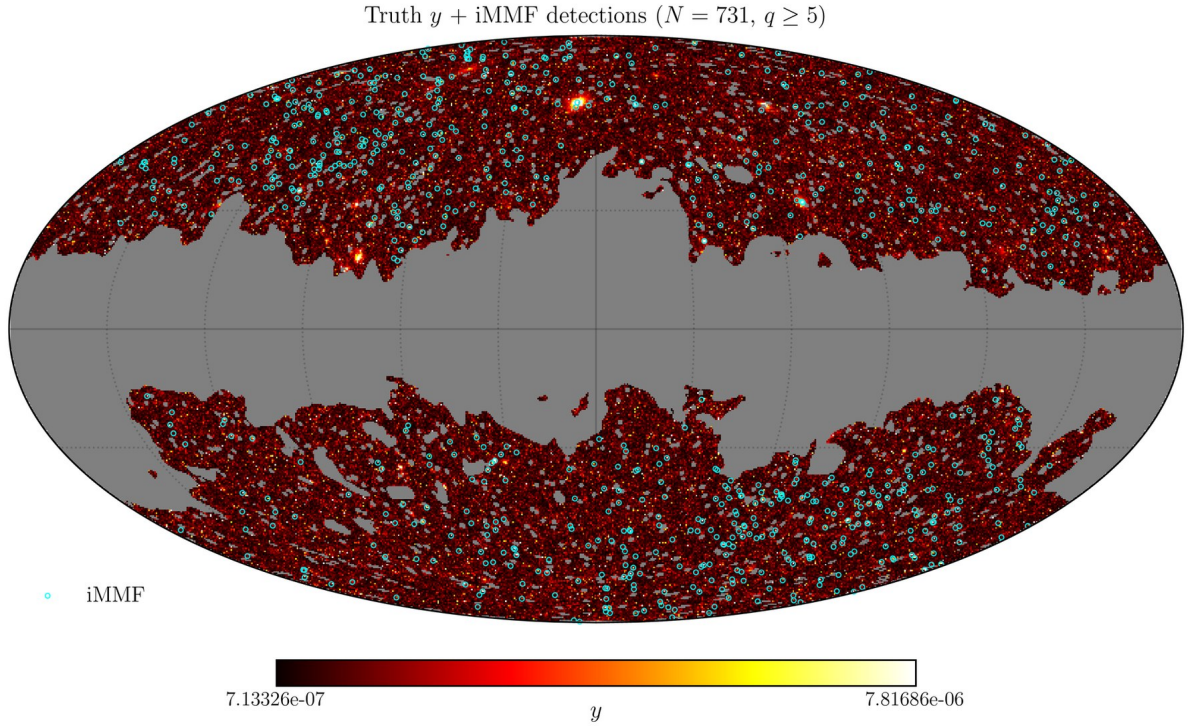


Figure 14: Truth Compton- y (PR4 GAL $\times$ PS footprint) with iMMF detections at $q_{\text{opt}} \geq 5$ (cyan circles scaled by θ_{500}).

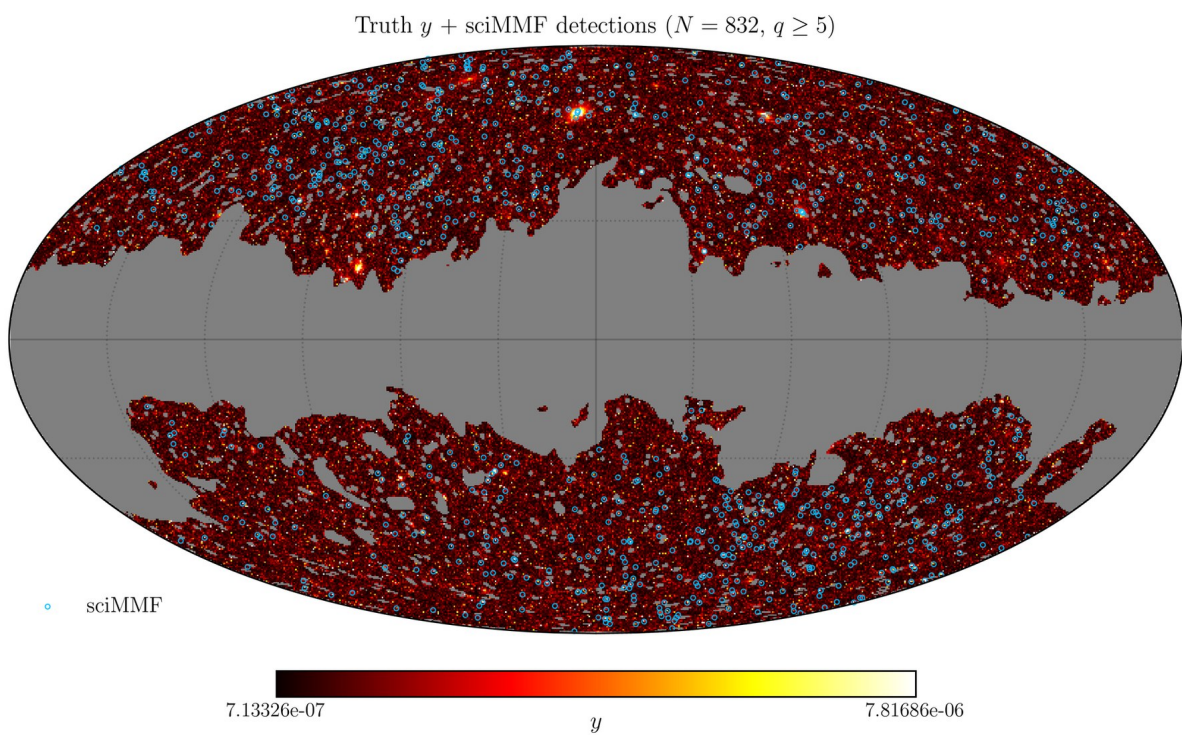


Figure 15: As Figure 14, for sciMMF detections.

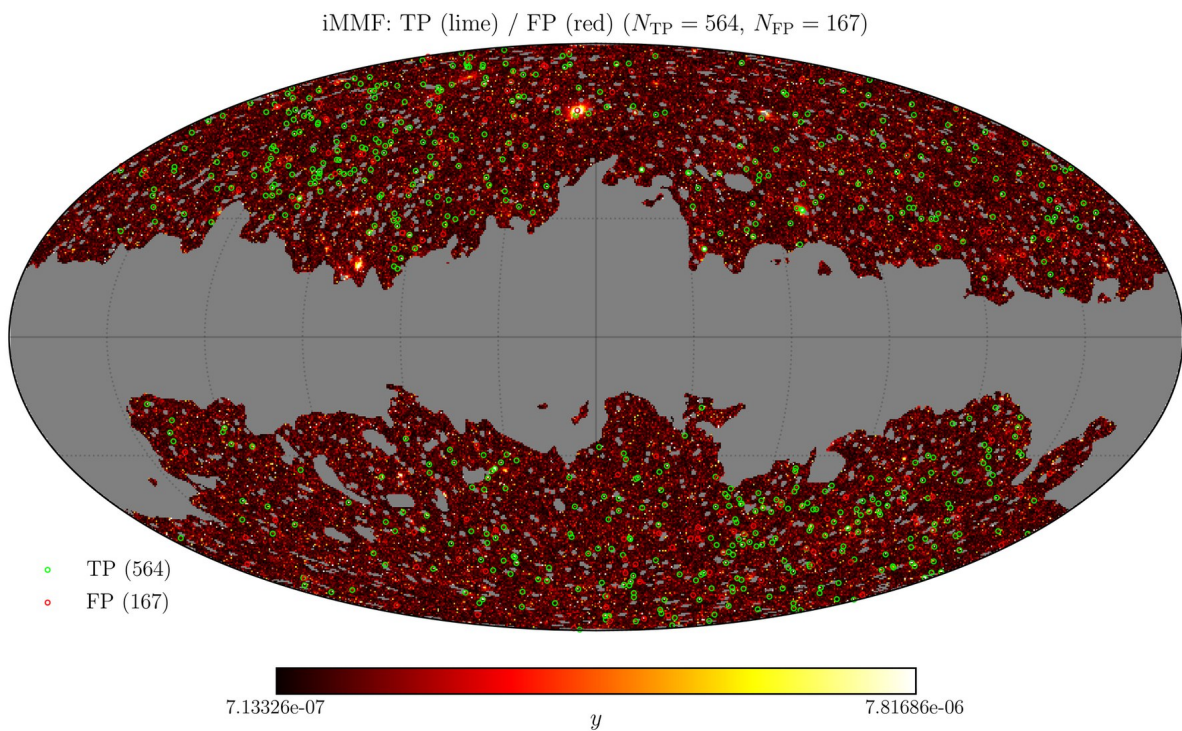


Figure 16: iMMF detections classified as true positives (lime) and false positives (red) after 10' matching to fixed-MMF $q_{\text{true}} \geq 5$ truth.

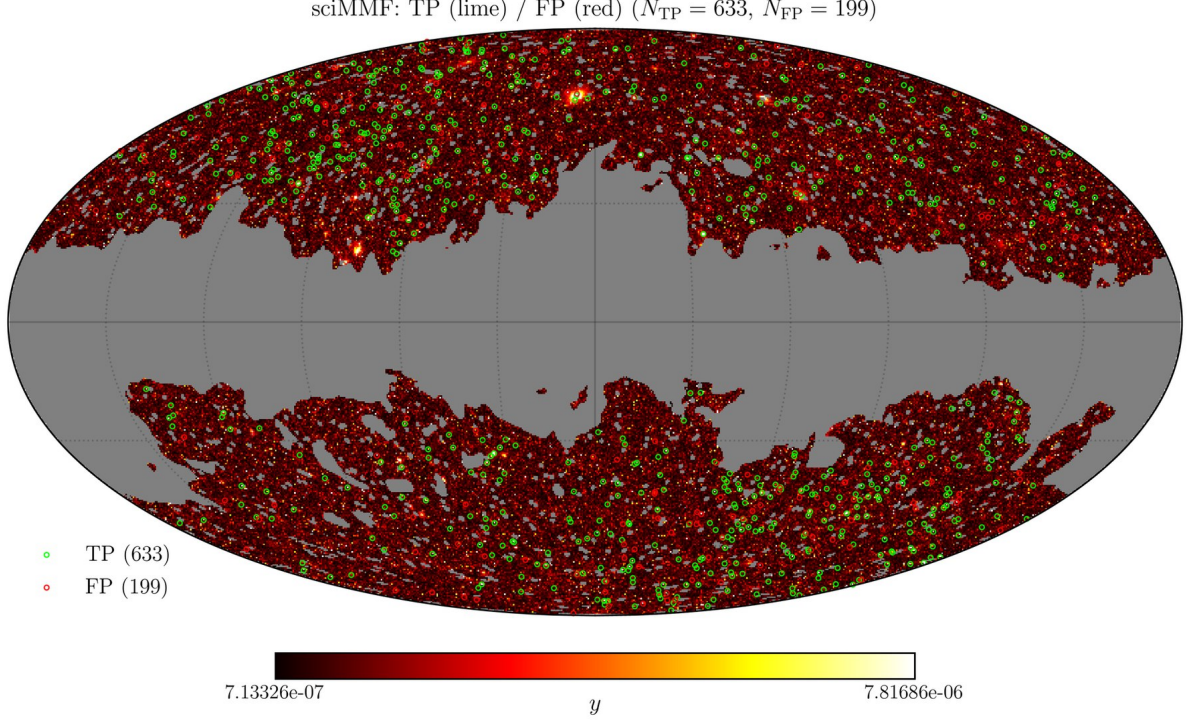


Figure 17: As Figure 16, for sciMMF.

truth SNR and purity versus detection threshold; Figure 20 compares the two methods. Completeness rises with truth SNR and approaches unity for the highest- q_{true} systems, while purity improves as the detection threshold is raised above $q = 5$.

Table 2: SZiFi footprint catalogues (NPIPE split A, $q_{\text{opt}} \geq 5$, PR4 GAL $\times$ PS, match radius $10'$, $z \leq 1$, truth = fixed-MMF $q_{\text{true}} \geq 5$). Completeness is $(N_{\text{truth}}^{\text{det.}} - N_{\text{undetected}})/N_{\text{truth}}^{\text{det.}}$.

Method	N_{det}	N_{TP}	N_{FP}	Purity	$N_{\text{truth}}^{\text{det.}}$	Completeness
iMMF	731	564	167	0.772	430	0.774
sciMMF	832	633	199	0.761	430	0.777

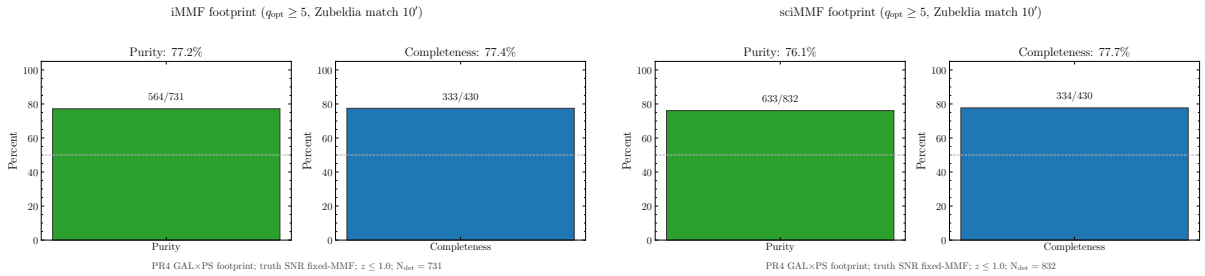


Figure 18: Purity and completeness for footprint catalogues at $q_{\text{opt}} \geq 5$: iMMF (left; 333/430 detectable truths found) and sciMMF (right; 334/430).

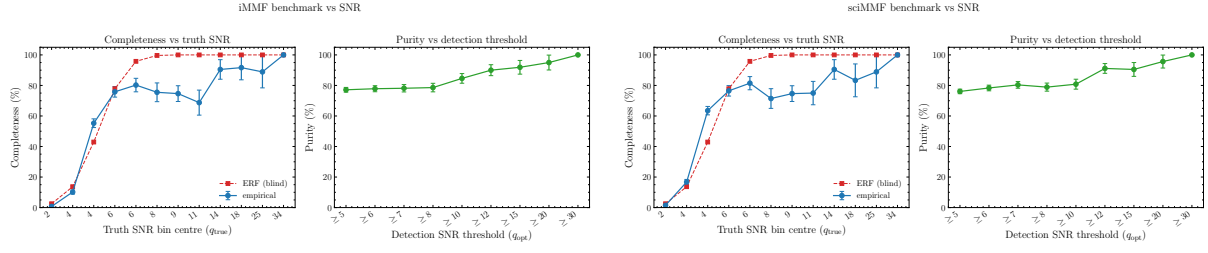


Figure 19: Completeness versus truth SNR bin and purity versus detection SNR threshold for iMMF (left) and sciMMF (right).

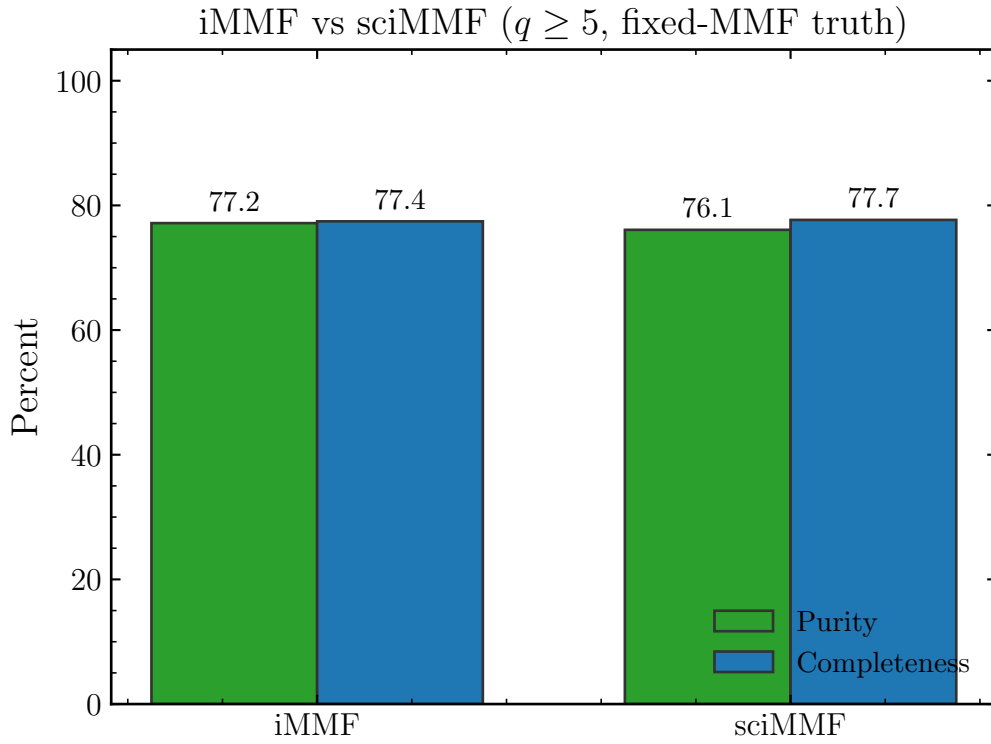


Figure 20: Purity and completeness for iMMF and sciMMF on the same footprint and truth definition.

5 Discussion and conclusions

We have constructed multi-frequency CMB skies from FLAMINGO lightcones and used them to validate secondary-anisotropy statistics and to test ILC y recovery and matched-filter cluster finding under Planck-like noise and masking. Our main conclusions are as follows.

1. **A four-component extragalactic sky is sufficient for controlled Planck-like SZ tests.** Lensed primary CMB, non-relativistic tSZ, kSZ and CIB, with common lightcone rotations, capture the secondaries and foregrounds most relevant to SZ science while excluding Galactic emission and radio sources by design. The resulting isolation is appropriate for algorithm tests; it is not a claim of full sky realism at 545–857 GHz.
2. **Spectral conversions must be applied consistently.** The non-relativistic tSZ function $f(x)$, the kSZ relation $\Delta T = -T_{\text{CMB}}b$, thermodynamic conversion of CIB intensity, and NPIPE unit conventions (K_{CMB} versus MJy sr^{-1}) fully specify the multi-frequency model used here.
3. **Native-resolution power spectra validate the maps.** CIB autos, frequency decorrelation, $\text{CIB} \times y$ and $\text{CIB} \times \kappa$ crosses, and $y/\text{kSZ}/\kappa$ autos recovered from the lightcone products are consistent with Yang et al. (2026). Downstream ILC and matched-filter analyses therefore start from spectrally faithful skies.
4. **ILC recovers large-scale y ; mild CIB deprojection helps, multi-moment constraints do not.** HILC and NILC recover the truth y spectrum after $5'$ beam deconvolution. In an eight-way constrained-ILC suite, CIB-only deprojection raises mid- ℓ correlation (ρ_{50-500} : $0.654 \rightarrow 0.671$) and transfer (T_{mid} : $0.909 \rightarrow 0.960$), whereas $\text{CIB} + \delta\beta + \delta T$ more than doubles σ_y and collapses the correlation with truth.
5. **iMMF and sciMMF recover clusters with purity and completeness both near 77%.** At $q_{\text{opt}} \geq 5$ on a Planck-like footprint we find purity $\approx 77.2\%$ (iMMF; 564 TP / 167 FP) and $\approx 76.1\%$ (sciMMF; 633 TP / 199 FP), with completeness of fixed-MMF-detectable systems $\approx 77\%$ (333/430 and 334/430 truths found). Sky maps of detections and TP/FP classifications show that high- q systems sit on bright y peaks while residual false positives include threshold noise and boundary matches. sciMMF yields a larger catalogue at similar purity.
6. **Limitations.** Results use a single lightcone, a single NPIPE realisation for cluster finding (split A), Gaussian beams, and no radio sources or Galactic dust. Completeness depends on the truth SNR definition. Multi-split cosmology analyses and fully redshift-matched halo comparisons remain for future work.

In summary, FLAMINGO-based multi-frequency skies provide a rigorous laboratory for millimetre component separation and cluster finding under realistic Planck noise and masking, with component-level truth that is unavailable in flight data.

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